HW 6

(6.2) Prove $0 < \delta < \pi$, helicity is left-handed

Prove $\pi < \delta < 2\pi$, helicity is right-handed

$$E(z,t) = |E_{eff}|\left[\hat{x}A\cos(kz - \omega t + \phi) + \hat{y}B\cos(kz - \omega t + \phi + \delta)\right]$$

$$E(0, \phi/\omega) = |E_{eff}|\left\{\hat{x}A\cos(0) + \hat{y}B\cos(\delta)\right\} = |E_{eff}|\left\{\hat{x}A + \hat{y}B\cos(\delta)\right\}$$

$k = \frac{2\pi}{\lambda}$

$$E(\lambda/4, \phi/\omega) = |E_{eff}|\left\{\hat{x}A\cos\left(\frac{2\pi}{\lambda}\frac{\lambda}{4}\right) + \hat{y}B\cos\left(\frac{2\pi}{\lambda}\frac{\lambda}{4} + \delta\right)\right\}$$

$$= |E_{eff}|\left\{\hat{y}B\cos\left(\frac{\pi}{2} + \delta\right)\right\}$$

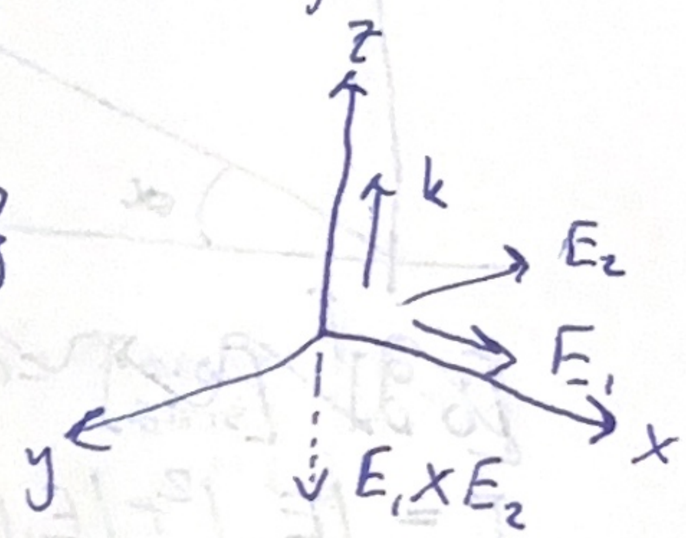
if $0 < \delta < \pi$ then $E_1(0, \phi/\omega) = |E_{eff}|\{\hat{x}A\}$

$\delta = \pi/2$ $\quad E_2(\lambda/4, \phi/\omega) = |E_{eff}|\{\hat{y}B\}$

if $\pi < \delta < 2\pi$

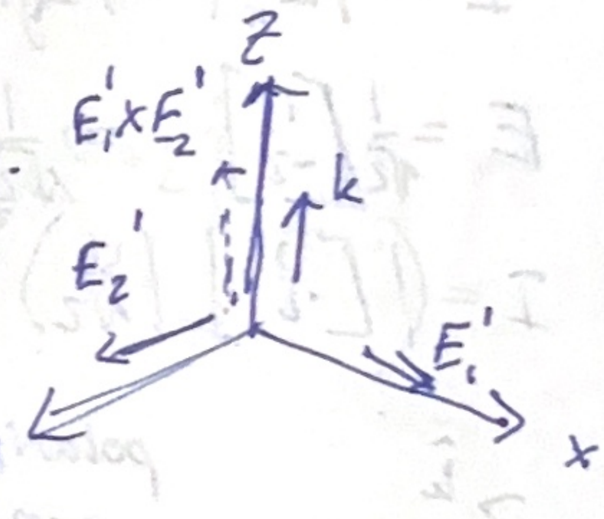
$\delta = \frac{3\pi}{2}$ then $E_1'(0, \phi/\omega) = |E_{eff}|\{\hat{x}A\}$

$E_2'(\lambda/4, \phi/\omega) = |E_{eff}|\{\hat{y}B\}$



since $E_1 \times E_2$ is in $-\hat{z}$ direction this implies left-handed helicity.

since $E_1' \times E_2'$ is in the $+\hat{z}$ direction this implies right-handed helicity.



(6.4) $\begin{bmatrix} A \\ Be^{i\delta} \end{bmatrix}$ orientation of major Axis? ellipticity of the light?

$$\tan(2\alpha) = \frac{2AB\cos(\delta)}{A^2 - B^2}$$

$$\sin(2\chi) = \frac{2AB\sin(\delta)}{A^2 + B^2}$$

$$e = |\tan(\chi)|$$

Case I: $A = B = \frac{1}{\sqrt{2}}$, $\delta = 0$

$\Rightarrow \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$ $\quad \tan(2\alpha) = \frac{1}{0} = \infty$

$2\alpha = \frac{\pi}{2} \Rightarrow \alpha = \frac{\pi}{4}$

$\sin(2\chi) = 0$

$2\chi = 0 \Rightarrow \chi = 0$ $\quad e = \tan(0) = 0$

$\boxed{\text{the ellipticity is 0, linearly polarized @ } \alpha = 45^\circ}$

case II: $A = B = \frac{1}{\sqrt{2}}$, $\delta = \frac{\pi}{2}$

$\tan(2\alpha) = \frac{2AB\cos(\delta)}{A^2 - B^2} = \frac{0}{0}$ the angle is undefined

$\sin(2\chi) = \frac{2AB\sin(\delta)}{A^2 + B^2} = \frac{1}{1} = 1 \quad 2\chi = \pi/2 \Rightarrow \chi = \frac{\pi}{4}$

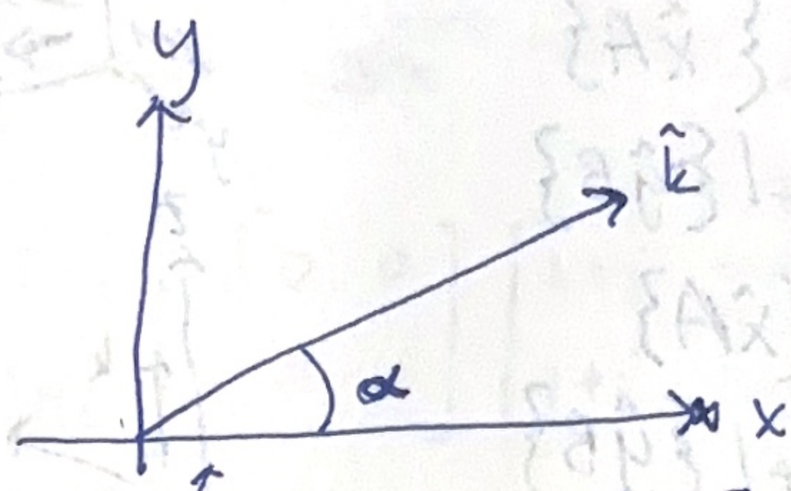
$e = |\tan(\pi/4)| = 1$ $\boxed{\text{Right circularly polarized light}}$

case III: $\tan(2\alpha) = \frac{2AB\cos(\delta)}{A^2 - B^2} = \frac{\frac{1}{\sqrt{2}}}{0} = \infty \Rightarrow \alpha = \frac{\pi}{4}$

$\sin(2\chi) = \frac{1/\sqrt{2}}{1} = \frac{1}{\sqrt{2}} = \chi = \frac{\pi}{8}$ $\quad e = |\tan(\chi)| = e = |\tan(\pi/8)| \approx 0.4$

elliptically polarized with angle 4

6.5 a.)



polarizer trans. along x $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

polarizer trans. along y $\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$

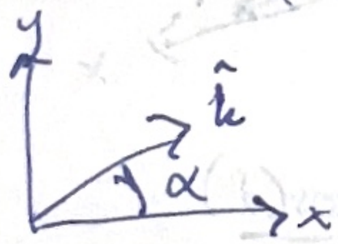
$\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\alpha \\ \sin\alpha \end{bmatrix} = \begin{bmatrix} 0 \\ \sin\alpha \end{bmatrix}$

$I = |E_x|^2 + |E_y|^2 = 0^2 + \sin^2\alpha \Rightarrow I = \sin^2\alpha$

b.) $E = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ -i \end{bmatrix} \Rightarrow \frac{1}{\sqrt{2}}\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 \\ -i \end{bmatrix} = \begin{bmatrix} 0 \\ -i \end{bmatrix}\frac{1}{\sqrt{2}}$

$I = \left(\begin{bmatrix} 0 \\ -i \end{bmatrix}\left(\frac{1}{\sqrt{2}}\right)\right) \Rightarrow \boxed{I = \frac{1}{2}}$

6.7



polarized @ θ from x-axis $\Rightarrow$

a.)

$\begin{bmatrix} \cos^2\theta & \cos\theta\sin\theta \\ \cos\theta\sin\theta & \sin^2\theta \end{bmatrix}\begin{bmatrix} \cos\alpha \\ \sin\alpha \end{bmatrix} = \begin{bmatrix} \cos^2\theta\cos\alpha + \cos\theta\sin\theta\sin\alpha \\ \cos\theta\sin\theta\cos\alpha + \sin^2\theta\sin\alpha \end{bmatrix} = \begin{bmatrix} \cos\theta(\cos\theta\cos\alpha + \sin\theta\sin\alpha) \\ \sin\theta(\cos\theta\cos\alpha + \sin\theta\sin\alpha) \end{bmatrix}$

$\begin{bmatrix} \cos^2\theta & \cos\theta\sin\theta \\ \cos\theta\sin\theta & \sin^2\theta \end{bmatrix}$

$\cos(A-B) = \cos(A)\cos(B) + \sin(A)\sin(B)$

$\hookrightarrow \begin{bmatrix} \cos\theta(\cos(\theta-\alpha)) \\ \sin\theta(\cos(\theta-\alpha)) \end{bmatrix} \Rightarrow I = \{\cos^2(\theta-\alpha)\cos^2\theta + \sin^2\theta\cos^2(\theta-\alpha)\}$

$\boxed{I = \cos^2(\theta-\alpha)}$

6.7 cont'd

b.) $E_i = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\-i\end{bmatrix} \Rightarrow \frac{1}{\sqrt{2}}\begin{bmatrix}\cos^2\theta & \cos\theta\sin\theta\\ \cos\theta\sin\theta & \sin^2\theta\end{bmatrix}\begin{bmatrix}1\\-i\end{bmatrix} = \begin{bmatrix}\cos^2\theta - i\cos\theta\sin\theta\\ \cos\theta\sin\theta - i\sin^2\theta\end{bmatrix}\frac{1}{\sqrt{2}}$

$\Rightarrow \frac{1}{\sqrt{2}}\begin{bmatrix}\cos\theta(\cos\theta - i\sin\theta)\\ \sin\theta(\cos\theta - i\sin\theta)\end{bmatrix} = \frac{e^{-i\theta}}{\sqrt{2}}\begin{bmatrix}\cos\theta\\ \sin\theta\end{bmatrix}$ $\quad I = \left(\frac{e^{-i2\theta}}{2}\right)(\cos^2\theta + \sin^2\theta)$

$\hookrightarrow I = e^{-i2\theta}/2 = \boxed{\frac{1}{2}}$

6.12

quarter λ fast axis @ 45° ; linear polarizer vertically ; quarter λ fast axis @ -45°

$\hookrightarrow \begin{bmatrix}\cos^2\theta + i\sin^2\theta & \sin\theta\cos\theta - i\sin\theta\cos\theta\\ \sin\theta\cos\theta - i\sin\theta\cos\theta & \sin^2\theta + i\cos^2\theta\end{bmatrix}$

$\frac{1}{2}\begin{bmatrix}1+i & 1-i\\ 1-i & 1+i\end{bmatrix}\begin{bmatrix}0 & 0\\ 0 & 1\end{bmatrix}\begin{bmatrix}1+i & -1+i\\ -1+i & 1+i\end{bmatrix}\frac{1}{2}$

$\frac{1}{4}\begin{bmatrix}0 & 0\\ 1-i & 1+i\end{bmatrix} \Rightarrow \frac{1}{4}\begin{bmatrix}1+i & -1+i\\ -1+i & 1+i\end{bmatrix}\begin{bmatrix}0 & 0\\ 1-i & 1+i\end{bmatrix}$

$= \frac{1}{4}\begin{bmatrix}0 + (-1+1+2i) & 0 + (-1-1)\\ 0 + (1+1) & 0 + (1-1+2i)\end{bmatrix} = \frac{1}{4}\begin{bmatrix}2i & -2\\ 2 & 2i\end{bmatrix} = \frac{1}{2}\begin{bmatrix}i & -1\\ 1 & i\end{bmatrix}$

RCP:

$\frac{1}{2}\begin{bmatrix}i & -1\\ 1 & i\end{bmatrix}\begin{bmatrix}1\\ -i\end{bmatrix} = \frac{1}{2}\begin{bmatrix}2i\\ 2\end{bmatrix} = \begin{bmatrix}i\\ 1\end{bmatrix}$ RCP pass

LCP: $\frac{1}{2}\begin{bmatrix}i & -1\\ 1 & i\end{bmatrix}\begin{bmatrix}1\\ i\end{bmatrix} = \begin{bmatrix}0\\ 0\end{bmatrix}$ LCP killed